MTI[redacted] Probability and[redacted]

(Lecture [redacted]) 6

[redacted] January 13, 2026

Probability Distribution of a Random Variable: So far we have introduced the concept of a random variable and noted that the concept of probability on the sample space was not used in this definition. In practice, however, random variables are of interest only when they are defined on a probability space. Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let X be a random variable defined on it.

Theorem 1: The random variable X defined on probability space $(\Omega, \mathcal{F}, P)$ induces a probability space $(\mathcal{R}, \mathcal{B}, Q)$ by means of the correspondence

$$Q(B) = P\left\{X^{-1}(B)\right\} = P\left\{\omega : X(w) \in B\right\} \ \ \forall \ \ B \in \mathcal{B}. \tag{1}$$

We write $Q = P(X^{-1})$ and call Q the (probability) distribution of X.

Proof: Clearly, $Q(B) \geq 0 \ \forall \ B \in \mathcal{B}$, and also $Q(\mathcal{R}) = P\left\{X \in \mathcal{R}\right\} = P(\Omega) = 1$. Let $B_i \in \mathcal{B}, \ \ i = 1, 2, ...,$ with $B_i \cap B_j = \phi, \ \ i \neq j$. Since the inverse image of a disjoint union of Borel sets is the disjoint union of their inverse images, we have

$$\begin{aligned} Q\left(\bigcup_{i=1}^{\infty} B_i\right) &= P\left\{X^{-1}\left(\bigcup_{i=1}^{\infty} B_i\right)\right\} \\ &= P\left\{\bigcup_{i=1}^{\infty} X^{-1}\left(B_i\right)\right\} \\ &= \sum_{i=1}^{\infty} P\left\{X^{-1}(B_i)\right\} \\ &= \sum_{i=1}^{\infty} Q(B_i). \end{aligned}$$

It follows that $(\mathcal{R}, \mathcal{B}, Q)$ is a probability space.

Cumulative Distribution Function: Let X be a random variable defined on $(\Omega, \mathcal{F}, P)$. Define a point function $F(.)$ on $\mathbb{R}$[redacted], namely

$$F(x) = Q(-\infty, x] = P\left\{\omega : X(\omega) \leq x\right\} \ \ \forall \ \ x \in \mathbb{R}.$$

The function F is called the *cumulative distribution function* of random variable X. We can simply write it as

$$F(x) = P\{X \leq x\}.$$

Properties of the Cumulative Distribution Function:

1. $0 \leq F(x) \leq 1$.

2. The distribution function F satisfies

$$F(-\infty) = \lim_{x \to -\infty} F(x) = 0 \text{ and } F(+\infty) = \lim_{x \to +\infty} F(x) = 1.$$

Proof: (i) Let $\{x_n\}$ be a decreasing sequence, such that, $\lim_{n\to\infty} x_n = -\infty$. Let

$$A_n = \{\omega : X(\omega) \leq x_n\}.$$

Then $\lim_{n\to\infty} A_n = \phi$. Now using continuity of probability measures

$$\lim_{n\to\infty} P(A_n) = P(\lim_{n\to\infty} A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right) = P(\phi) = 0.$$

$$\Rightarrow \lim_{n\to\infty} F(x_n) = 0.$$

$$\Rightarrow \lim_{x\to-\infty} F(x) = 0.$$

(ii) Let $\{x_n\}$ be an increasing sequence, such that, $\lim_{n\to\infty} x_n = +\infty$. Let

$$A_n = \{\omega : X(\omega) \leq x_n\}.$$

Then $\lim_{n\to\infty} A_n = \Omega$. Using continuity of probability measures

$$\lim_{n\to\infty} P(A_n) = P(\lim_{n\to\infty} A_n) = P\left(\bigcup_{n=1}^{\infty} A_n\right) = P(\Omega) = 1.$$

$$\Rightarrow \lim_{n\to+\infty} F(x_n) = 1.$$

$$\Rightarrow \lim_{x\to+\infty} F(x) = \bullet 1$$

3. F is nondecreasing, i.e., if $x < y$, then $F(x) \leq F(y)$.

Proof: If $x < y$, then

$$\{\omega : X(\omega) \leq x\} \subseteq \{\omega : X(\omega) \leq y\}$$

$$\Rightarrow P\{\omega : X(\omega) \leq x\} \leq \{\omega : X(\omega) \leq y\}.$$

$$\Rightarrow F(x) \leq F(y).$$

4. F is right continuous.

   Proof: Let $\{x_n\}$ be a decreasing sequence, such that, $\lim_{n\to\infty} x_n = x$. Let

$$A_n = \{\omega : X(\omega) \leq x_n\}.$$

   Now

$$\lim_{n\to\infty} P(A_n) = P(\lim_{n\to\infty} A_n)$$
$$\Rightarrow \lim_{n\to\infty} \{F(x_n) - F(x)\} = 0.$$
$$\Rightarrow \lim_{n\to\infty} F(x_n) = F(x).$$

   Since $\{x_n\}$ is a decreasing sequence to x (obviously from right), where $\{x_n\}$ is any arbitrary sequence, we can have

$$\Rightarrow \lim_{\epsilon\to 0} F(x+\epsilon) = F(x).$$

Remark 1: The aforementioned properties are satisfied by all cumulative distribution functions. Also, any function that satisfies these properties is necessarily a cumulative distribution function for some random variable X.

Theorem 2: The set of discontinuity points of a cumulative distribution function is at most countable.

Proof: Let a distribution function $F(x)$ has n points of discontinuities in a finite interval $[a, b]$, such that

$$A = \{x_1, x_2, ..., x_n\}.$$

Since $A \subseteq [a, b]$,

$$P(A) \leq F(b) - F(a) \text{ or } \sum_{i=1}^{n} P(x_i) \leq F(b) - F(a).$$

and it follows that number of points x in $[a, b]$ with jump $p(x) > \epsilon > 0$ is at most $\epsilon^{-1}\{F(b) - F(a)\}$. Thus for every integer N, the number of discontinuity points with jump greater than $1/N$ is finite. It follows that there are no more than a countable number of discontinuity points in every finite interval $[a, b]$.

Discrete Random Variable: A random variable X defined on $(\Omega, \mathcal{F}, P)$ is said to be of discrete type, if there exists a countable set $E \subseteq \mathcal{R}$ such that $P(X \in E) = 1$. In simple terms, if a random variable X takes at most a countable number of values or countably infinite number of values, it

is called a *discrete random variable*. In other words, a real valued function defined on a discrete sample space is called a discrete random variable.

Probability Mass Function: Let X is a discrete random variable taking at most a countable infinite number of values $x_1, x_2, ...$ with probability p_i $(i = 1, 2, ...)$. We have

$$P(X = x_i) = p_i, \;\; i = 1, 2, ..., \;\; p_i \geq 0 \;\; \forall \;\; i.$$

Then $\sum\limits_{i=1}^{\infty} p_i = 1$. The collection of numbers $\{p_i\}$ satisfying $P(X = x_i) = p_i \geq 0 \;\; \forall \;\; i$, and $\sum\limits_{i=1}^{\infty} p_i = 1$, is called the *probability mass function* (PMF) of X.

The distribution function F of X with PMF, p_i is given by

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} p_i.$$

Result: $p_j = P(X = x_j) = F(x_j) - F(x_{j-1}), j = 1, 2, ...$ where F is the distribution function of a random variable X.

Proof: Let $x_1 < x_2 < ...$, we have

$$\begin{aligned} F(x_j) &= P(X \leq x_j) \\ &= \sum_{i=1}^{j} P(X = x_i) \\ &= \sum_{i=1}^{j} p_i, \end{aligned}$$

and

$$\begin{aligned} F(x_{j-1}) &= P(X \leq x_{j-1}) \\ &= \sum_{i=1}^{j-1} P(X = x_i) \\ &= \sum_{i=1}^{j-1} p_i. \end{aligned}$$

This implies that

$$F(x_j) - F(x_{j-1}) = p_j.$$

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Example 1: The simplest example of a discrete random variable X is 'X degenerate at c', $P(X = c) = 1$. The corresponding CDF is given by

$$F(x) = \begin{cases} 0, & x < c, \\ 1, & x \geq c. \end{cases}$$

Example 2: A box contains good and defective items. If an item drawn is good, we assign the number 1 to the drawing; otherwise, the number 0. Let p be the probability of drawing a good item at random. Then $P(X = 0) = 1 - p, P(X = 1) = p$, and the corresponding CDF is given by

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \leq x < 1, \\ 1, & 1 \leq x. \end{cases}$$

Example 3: A random variable X has the following PMF:

x	0	1	2	3	4	5	6	7
$p(x)$	0	a	$2a$	$2a$	$3a$	a^2	$2a^2$	$7a^2 + a$

Find the value of 'a' and then obtain the CDF of X.

Solution: Since $\sum_{x=0}^{7} p(x) = 1$. One may easily obtain $a = 1/10$. The CDF is given by

$$F(x) = P(X \leq x) = \begin{cases} 0, & x < 1, \\ 1/10, & 1 \leq x < 2, \\ 3/10, & 2 \leq x < 3, \\ 5/10, & 3 \leq x < 4, \\ 8/10, & 4 \leq x < 5, \\ 81/100, & 5 \leq x < 6, \\ 83/100, & 6 \leq x < 7, \\ 1, & x \geq 7. \end{cases}$$

Clearly, in this example, the graph of CDF will look like a *step function* having jump $p(x_i)$ at x_i $(i = 1, 2, ..., 7)$, and being constant between each pair of integers.

Continuous Random Variable and Probability Density Function: Let X be a random variable defined on $(\Omega, \mathcal{F}, P)$ with CDF F. Then X is said to be of continuous type if F is absolutely continuous, i.e., if there exists a nonnegative continuous function $f(x)$ such that for every real number x we have

$$F(x) = \int_{-\infty}^{x} f(t)dt.$$

The function f is called the *probability density function* (PDF) of the random variable X. The PDF f has the following properties:

1. $f \geq 0$.
2. $\lim_{x \to +\infty} F(x) = F(+\infty) = \int_{-\infty}^{\infty} f(t)dt = 1$.
3. Let a and b be any two real numbers with $a < b$, Then

$$P(a < X \leq b) = F(b) - F(a) = \int_{a}^{b} f(t)dt.$$

Remark 1: Let X be a continuous random variable with PDF f. Then for every Borel set $B \in \mathcal{B}$,

$$P(B) = \int_{B} f(t)dt.$$

Remark 2: If F is absolutely continuous and f is continuous at x, we have

$$F'(x) = \frac{dF(x)}{dx} = f(x).$$

Remark 3: Every nonnegative real function f that is integrable over $\mathbb{R}$ and satisfies

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

is the PDF of some continuous random variable X.

Remark 4: In the discrete case, $P(X = a)$ is the probability that X takes the value a. In the continuous case, $f(a)$ is not the probability that X takes the value a. Indeed, if X is of the continuous type, it assumes every value with probability 0.

106 Stochastic Processes

MTL[redacted] Probability and [redacted]

(Lecture 5)

[redacted] January 10, 2026

Independence of Events: Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $A, B \in \mathcal{F}$, with $P(B) > 0$. By the multiplication rule we have

$$P(A \cap B) = P(B)P(A|B).$$

In many experiments the information provided by B does not affect the probability of event A, i.e., $P(A|B) = P(A)$. Eg., winning the toss by the Indian cricket team captain in consecutive matches are independent events. Whereas, let event A be the life of an equipment exceeding 100 hours and event B be life of the equipment exceeding 200 hours. Then events A and B are dependent events.

Two events, A and B are said to be independent if and only if

$$P(A \cap B) = P(A)P(B).$$

Remark 1: We have not placed any restriction on $P(A)$ or $P(B)$. Thus conditional probability is not defined when $P(A)$ or $P(B) = 0$, but independence is. Clearly, if $P(A) = 0$, then A is independent of every $E \in \mathcal{F}$. Also, any event $A \in \mathcal{F}$ is independent of ϕ and Ω.

Theorem 1: If A and B are independent events, then

$$P(A|B) = P(A), \text{ if } P(B) > 0,$$

$$P(B|A) = P(B), \text{ if } P(A) > 0.$$

Theorem 2: If A and B are independent events, so are A and B^c, A^c and B, and A^c and B^c.

Proof: The proof of all three cases is provided below:

$$\begin{aligned} P(A \cap B^c) &= P(A - (A \cap B)) \\ &= P(A) - P(A \cap B) \ \ [\text{since}(A \cap B) \subseteq A] \\ &= P(A)[1 - P(B)] \\ &= P(A)P(B^c). \end{aligned}$$

$$
\begin{aligned}
P(A^c \cap B) &= P(B - (A \cap B)) \\
&= P(B) - P(A \cap B) \ \ [\text{since}(A \cap B) \subseteq B] \\
&= P(B)[1 - P(A)] \\
&= P(A^c)P(B).
\end{aligned}
$$

$$
\begin{aligned}
P(A^c \cap B^c) &= 1 - P(A \cup B) \ \ [\text{De Morgan's Law}] \\
&= 1 - [P(A) + P(B) - P(A \cap B)] \ \ [\text{Law of Addition}] \\
&= 1 - P(A) - P(B) + P(A)P(B) \ \ [\text{A and B are independent}] \\
&= [1 - P(A)] - P(B)[1 - P(A)] \\
&= [1 - P(A)][1 - P(B)] \\
&= P(A^c)P(B^c).
\end{aligned}
$$

Remark 2: If two events (each with nonzero probability) are mutually exclusive, they are not independent since the occurence of one will automatically preclude the occurence of the other. Similarly if two events are independent (each with nonzero probability), then they can't be mutually exclusive.

Remark 2: Three events A, B, C are independent, if

$$P(A \cap B) = P(A)P(B),$$

$$P(A \cap C) = P(A)P(C),$$

$$P(B \cap C) = P(B)P(C),$$

$$P(A \cap B \cap C) = P(A)P(B)P(C).$$

Remark 3: "n" Events $A_1, A_2, ..., A_n$ are independent, if

$$P(A_i \cap A_j) = P(A_i)P(A_j), \ \ i \neq j; \ \ i, j = 1, 2, ..., n,$$

$$P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k), \ \ i \neq j \neq k; \ \ i, j, k = 1, 2, ..., n,$$

$$\vdots$$

$$P(A_1 \cap A_2 \cap ... \cap A_n) = P(A_1)P(A_2)...P(A_n).$$

The total number of conditions to determine the independence of "n" events is $2^n - n - 1$.

Remark 4: Independence implies pairwise independence, but the converse is not true.

Example 1: Two machines, A and B operate independently. The probability that machine A fails on a given day is 0.01, and the probability that machine B fails on a given day is 0.02. Let E_1 be the event that machine A fails on a given day, and E_2 be the event that machine B fails on the same day. Determine if events E_1 and E_2 are independent.

Solution: We have $P(E_1) = 0.01$ and $P(E_2) = 0.02$, and $P(E_1 \cap E_2) = P(E_1)P(E_2) = 0.01 \times 0.02 = 0.0002$. Hence E_1 and E_2 are independent.

Example 2: In a "Probability and Statistics" course exam, Student 'A' can solve 75% of the problems and Student 'B' can solve 70% of the problems. Both the students solve the problems independently. What is the probability that either A or B can solve a randomly selected problem?

Solution: $P(A) = 0.75, P(B) = 0.70, P(A \cap B) = P(A)P(B) = 0.75 \times 0.70 = 0.525$. We have to find $P(A \cup B)$, which is given by $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B) = 0.75 + 070 - 0.525 = 0.925$.

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Random Variable: Have a look at the following scenarios:

1. A bank may like to know the number of fraudulent transactions than the actual sequence of occurrence of fraudulent and genuine transactions.

2. One might be interested in knowing how many matches India won in the last twenty test matches against Australia rather than a sequence of wins and defeats.

3. An E-commerce website may like to know the number of repeat orders rather than a sequence of orders.

4. A professor may like to know that how many classes were missed by a student rather than the sequence of presence and absence.

5. A social media influencer may like to know the total number of likes in his/her last 100 instagram reels rather than a sequence of likes and dislikes.

What is a special observation in the aforementioned scenarios? The answer is, in all these cases, we are not interested in the sequence of all the outcomes of a random experiment. Rather we are interested in knowing that how many times a particular outcome appears. The numerical measure of a particular outcome in a random experiment is known as a *random variable*.

Let $(\Omega, \mathcal{F})$ be a measurable space. A finite, single-valued function X that maps Ω into $\mathbb{R}$ (i.e., $X : \Omega \to \mathbb{R}$) is called a random variable if the inverse images under X of all Borel sets in $\mathbb{R}$ are events, i.e.,

$$X^{-1}(B) = \{\omega : X(w) \in B\} \in \mathcal{F} \ \forall \ B \in \mathcal{B}. \tag{1}$$

To verify whether a real-valued function on $(\Omega, \mathcal{F})$ is a random variable, it is not necessary to check that (1) holds for all Borel sets $B \in \mathcal{B}$. It suffices to verify (1) for any class $\mathcal{C}$ of subsets of $\mathbb{R}$ that generates $\mathcal{B}$. By taking $\mathcal{C}$ to be the class of semiclosed intervals $(-\infty, x], x \in \mathbb{R}$, we get the following result:

Result: X is a random variable if and only if for each $x \in \mathbb{R}$,

$$\{\omega : X(w) \leq x\} = \{X \leq x\} \in \mathcal{F}. \tag{2}$$

Example 1: Let $\Omega = \{\text{W,L}\}$, and $\mathcal{F}$ be the class of all subsets of Ω. Here W and L denote *win* and *defeat* of a team in a match. Define X by

$$X(W) = 1, X(L) = 0.$$

Then

$$X^{-1}(-\infty, x] = \{\omega : X(w) \leq x\} = \begin{cases} \phi, & x < 0, \\ \{\text{L}\}, & 0 \leq x < 1, \\ \Omega, & x \geq 1. \end{cases}$$

Here $\mathcal{F} = \mathcal{P}(\Omega)$ (power set of the sample space), and $X^{-1}(-\infty, x] = \{\omega : X(w) \leq x\} \in \mathcal{F}$. Thus X is a random variable.

Example 2: Let $\Omega = \{\text{WW,WL,LW,LL}\}$, and $\mathcal{F}$ be the class of all subsets of Ω. Here W and L denote *win* and *defeat* of a team in two consecutive matches respectively. Define X by

$$X(\omega) = \text{Number of W's in } \omega.$$

Then $X(\text{WW}) = 2,\ X(\text{WL}) = X(\text{LW}) = 1,\ X(\text{LL}) = 0.$

$$X^{-1}(-\infty, x] = \{\omega : X(w) \leq x\} = \begin{cases} \phi, & x < 0, \\ \{\text{LL}\}, & 0 \leq x < 1, \\ \{\text{LL, LW, WL}\}, & 1 \leq x < 2, \\ \Omega, & x \geq 2. \end{cases}$$

Here $\mathcal{F} = \mathcal{P}(\Omega)$, and $X^{-1}(-\infty, x] = \{\omega : X(w) \leq x\} \in \mathcal{F}$. Thus X is a random variable.

Example 3: For any set $A \subseteq \Omega$, define

$$I_A(\omega) = \begin{cases} 0, & \omega \notin A, \\ 1, & \omega \in A. \end{cases}$$

$I_A(\omega)$ is called the indicator function of set A. I_A is a random variable if and only if $A \in \mathcal{F}$.

Remark 1: The notion of probability does not enter into the definition of a random variable.

Remark 2: Let $(\Omega, \mathcal{F})$ be a discrete sample space, i.e., Ω be a countable set of points and $\mathcal{F}$ be the class of all subsets of Ω. Then every numerical-valued function defined on $(\Omega, \mathcal{F})$ is a random variable.

Remark 3: Let X be a random variable defined on $(\Omega, \mathcal{F})$ and let a, b be constants. Then $aX+b$ is also a random variable on $(\Omega, \mathcal{F})$. Moreover, X^2 is a random variable and so also is $1/X$, provided that $\{X = 0\} = \phi$.

MTL106: Stochastic Processes

[redacted] **Probability and** [redacted]

(Lecture 4)

[redacted] January 9, 2026

Conditional Probability: So far, we have discussed about the probabilities of events based on the assumption that no information was available about the random experiment other than the sample space. Sometimes, however, it is known that an event E_2 has occurred. The quetion is, can we use this information in making a statement about the outcome of another event E_1? How?

Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $E_2 \in \mathcal{F}$ with $P(E_2) > 0$. Then, for an arbitrary $E_1 \in \mathcal{F}$ we shall write

$$P(E_1|E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)}, \tag{1}$$

and the quantity so defined is known as the *conditional probability* of E_1, given E_2.

Remark 1: Conditional probability remains undefined when $P(E_2) = 0$.

Theorem 1: Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $H \in \mathcal{F}$ with $P(H) > 0$. Then, $(\Omega, \mathcal{F}, P_H)$, where $P_H(A) = P\{A|H\} \ \forall \ A \in \mathcal{F}$, is a probability space.

Proof: Clearly, $P_H(A) = P\{A|H\} \geq 0, \ \forall \ A \in \mathcal{F}$. Also,

$$P_H(\Omega) = \frac{P(\Omega \cap H)}{P(H)} = 1.$$

If $A_1, A_2, ...$ is a disjoint sequence of sets in $\mathcal{F}$, then

$$\begin{aligned} P\left(\bigcup_{i=1}^{\infty} A_i|H\right) &= \frac{P\left\{\bigcup_{i=1}^{\infty} A_i \cap H\right\}}{P(H)} \\ &= \frac{P\left\{\bigcup_{i=1}^{\infty} (A_i \cap H)\right\}}{P(H)}. \end{aligned} \tag{2}$$

Since A_i are disjoint, $A_i \cap H$ are also disjoint. Thus we shall have

$$\begin{aligned} \frac{P\left\{\bigcup_{i=1}^{\infty} (A_i \cap H)\right\}}{P(H)} &= \sum_{i=1}^{\infty} \frac{P(A_i \cap H)}{P(H)} \\ &= \sum_{i=1}^{\infty} P(A_i|H). \end{aligned} \tag{3}$$

Substituting (3) in (2), we obtain

$$P\left(\bigcup_{i=1}^{\infty} A_i|H\right) = \sum_{i=1}^{\infty} P(A_i|H).$$

One can observe that $P_H(A)$ is a probability measure on $(\Omega, \mathcal{F})$ as it satisfies all three axioms of a probability function. Also $(\Omega, \mathcal{F}, P_H)$ will be a probability space.

Properties of Conditional Probability:

1. **Multiplication Rule:** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A, B \in \mathcal{F}$, with $P(A) > 0, P(B) > 0$. Then

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B).$$

2. **General Multiplication Rule:** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A_1, A_2, ..., A_n \in \mathcal{F}$, with $P\left(\bigcap_{i=1}^{n-1} A_i\right) > 0$. Then

$$P\left(\bigcap_{i=1}^{n} A_i\right) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)...P\left\{A_n \middle| \bigcap_{i=1}^{n-1} A_i\right\}.$$

3. **Law of Total Probability:** Let $A \in \mathcal{F}$ and let $H_i, i = 1, 2, ...$ be events that partition Ω, $\left(\text{i.e., } \bigcup_{i=1}^{\infty} H_i = \Omega \text{ and } H_i \cap H_j = \phi \ \forall \ i \neq j\right)$ with $P(H_i) > 0, \ \forall \ i$. Then,

$$P(A) = \sum_{i=1}^{\infty} P(A|H_i)P(H_i).$$

Proof: Since $H_i, i = 1, 2, ...$ partition Ω, thus $A \cap H_i$ partition A. Hence, using the axiom of countable additivity, we have

$$P(A) = P\left\{\bigcup_{i=1}^{\infty}(A \cap H_i)\right\} = \sum_{i=1}^{\infty} P(A \cap H_i),$$

and

$$P(A \cap H_i) = P(A|H_i)P(H_i).$$

This implies

$$\sum_{i=1}^{\infty} P(A \cap H_i) = \sum_{i=1}^{\infty} P(A|H_i)P(H_i).$$

The result now follows on combining above expressions.

Note: In particular, if H is such that $0 < P(H) < 1$, we have

$$P(A) = P(A|H)P(H) + P(A|H^c)P(H^c).$$

4. **Bayes Theorem:** Let H_i be a disjoint sequence of events such that $P(H_i) > 0, i = 1, 2, ...,$ and $\bigcup_{i=1}^{\infty} H_i = \Omega$. Let $B \in \mathcal{F}$ with $P(B) > 0$. Then

$$P(H_i|B) = \frac{P(H_i)P(B|H_i)}{\sum_{i=1}^{\infty} P(H_i)P(B|H_i)}, \quad i = 1, 2, ...$$

Proof: Using multiplication rule, we have

$$P(B \cap H_i) = P(B)P(H_i|B) = P(H_i)P(B|H_i),$$

and it follows that

$$P(H_i|B) = \frac{P(H_i)P(B|H_i)}{P(B)}.$$

Using the law of total probability, it can be written as

$$P(H_i|B) = \frac{P(H_i)P(B|H_i)}{\sum_{i=1}^{\infty} P(H_i)P(B|H_i)},$$

and the theorem follows.

Remark 1: Let $H_1, H_2, ...$ are all the "causes" that lead to the outcome of a random experiment. Let H_i be the set of outcomes corresponding to the ith cause. Assume that the probabilities $P(H_i), i = 1, 2, ...$ called the *prior probabilities*, can be assigned. Now suppose that the experiment results in an event B of positive probability. This information leads to a reassessment of the prior probabilities. The conditional probabilities $P(H_i|B)$ are called the *posterior probabilities*. The Bayes rule can be interpreted as a rule giving the probability that the observed event B was due to cause H_j.

Example 1: An agency "A" reports on-time performance for airlines at major Indian airports. Air India, Indigo, and Spicejet share terminal 3 at IGI airport Delhi. Suppose that the percentage of on-time flights reported was 76.8% for Air India, 71.5% for Indigo, and 82.2% for Spicejet. Assume that 30% of the flights arriving at terminal 3 are Air India flights, 32% are Indigo flights, and 38% are Spicejet flights. An announcement is made that Flight 1382 will be arriving at gate no. 20 of terminal 3. What is the probability that Flight 1382 will arrive on time?

Solution: Let O = flight arrives on time

L = flight arrives late

A = Air India flight

I = Indigo flight

S = Spicejet flight

Given that

$$P(O|A) = 0.768, P(O|I) = 0.715, P(O|S) = 0.822,$$

$$P(A) = 0.30, P(I) = 0.32, P(S) = 0.38.$$

Using multiplication law, we have

$$P(A \cap O) = P(O|A)P(A) = 0.768 \times 0.30 = 0.2304,$$

$$P(I \cap O) = P(O|I)P(I) = 0.715 \times 0.32 = 0.2288,$$

$$P(S \cap O) = P(O|S)P(S) = 0.822 \times 0.38 = 0.3124.$$

Now the probability of an on-time flight is given by

$$P(O) = 0.2304 + 0.2288 + 0.3124 = 0.7716.$$

Example 2: Black boxes used in aircrafts are manufactured by three companies A, B and C. 75% are manufactured by A, 15% by B, and 10% by C. The defect rates of black boxes manufactured by A, B, and C are 4%, 6%, and 8%, respectively. If a black box tested randomly is found to be defective, what is the probability that it is manufactured by company A?

Solution: Let $P(A), P(B), P(C)$ be the events corresponding to the black box being manufactured by companies A, B, and C, respectively, and $P(D)$ be the probability of a defective black box. We are interested in calculating the probability $P(A|D)$, which is given by

$$P(A|D) = \frac{P(D|A)P(A)}{P(D)}.$$

Now we have $P(A) = 0.75, P(D|A) = 0.04$, and $P(D)$ is given by

$$\begin{aligned} P(D) &= P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C) \\ &= 0.75 \times 0.04 + 0.15 \times 0.06 + 0.10 \times 0.08 \\ &= 0.047. \end{aligned}$$

This implies that $P(A|D) = (0.04)(0.75)/(0.047) = 0.6382$.

MTL106: Probability and Stochastic Processes

January 7, 2026

(Lecture 3)

[redacted]

Remarks 1: If Ω is discrete and contains at most $n(<\infty)$ points, each single point set $\{\omega_j\}$, $j = 1, 2, ...n$, is an elementary event, and it is sufficient to assign probability to each $\{\omega_j\}$. . Then if $E \in \mathcal{F}$, where $\mathcal{F}$ is the class of all subsets of $\Omega, P(E) = \sum_{w \in E} P\{\omega\}$. One such assignment is the *equally likely* assignment or the assignment of uniform probabilities. According to this assignment, $P\{\omega_j\} = 1/n$, $j = 1, 2, ..., n$. Thus $P(E) = m/n$ if E contains m elementary events, $1 \leq m \leq n$.

Remark 2: If Ω is discrete and contains a countable number of points, one can't make an equally likely assignment of probabilities. It suffices to make the assignment for each elementary event. If $E \in \mathcal{F}$, where $\mathcal{F}$ is the class of all subsets of Ω, define $P(E) = \sum_{w \in E} P\{\omega\}$.

Remark 3: If Ω contains uncountably many points, each one-point set is an elementary event, and again one cannot make an equally likely assignment of probabilities. Indeed, one cannot assign positive probability to each elementary event without violating the axiom $P(\Omega) = 1$. In this case one assigns probabilities to compound events consisting of intervals. For example, if $\Omega = [0, 1]$ and $\mathcal{F}$ is the Borel σ-field of all subsets of Ω, the assignment $P[I] =$ length of I, where I is a sub-interval of Ω, defines a probability.

Example 1: Let $\Omega = \{0, 1\}$ and $\mathcal{F}$ is the σ-field of all subsets of Ω. Suppose we define P on $\mathcal{F}$ as follows:

$$P\{1\} = 1/2, \;\; P\{0\} = 1/2.$$

Then P clearly defines a probability. Similarly, $P\{1\} = 2/3$, $P\{0\} = 1/3$, and $P\{1\} = 1$, $P\{0\} = 0$, are probabilities defined on $\mathcal{F}$. Indeed, $P\{1\} = p$, $P\{0\} = 1 - p$, $(0 < p < 1)$ defines a probability on $(\Omega, \mathcal{F})$.

Example 2: Let $\Omega = \{1, 2, 3, ...\}$ be the set of positive integers, and let $\mathcal{F}$ be the class of all subsets of Ω. Suppose we define P on $\mathcal{F}$ as follows:

$$P\{i\} = 1/2^i, \;\; i = 1, 2, ...$$

Then $\sum_{i=1}^{\infty} P(i) = \sum_{i=1}^{\infty} 1/2^i = 1$, and P defines a probability.

Example 3: Let $\Omega = \{0, \infty\}$, and let $\mathcal{F} = \mathcal{B}$ be the Borel σ-field on Ω. Suppose we define P on $\mathcal{B}$ as follows:

For each interval $I \subseteq \Omega$,

$$P\{I\} = \int_I e^{-x} dx.$$

Then $P(I) \geq 0$, $P(\Omega) = 1$, and P is countably additive by properties of integrals. Clearly, P proves to be a probability function.

Some Important Properties of the Probability Function:

1. $P(\phi) = 0$.

   Proof. Let $E_1 = \Omega$, $E_2 = E_3 = ... = \phi$. Then using Axiom (ii), we have

   $$\begin{aligned} P(\Omega) &= P(E_1) + P(E_2) + P(E_3) + ... \\ &= P(\Omega) + P(\phi) + P(\phi) + ... \end{aligned}$$

   Since $P(\Omega) = 1$ [from Axiom (iii)], Axiom (i) implies that $P(\phi) \geq 0$, we must have $P(\phi) = 0$.

2. For any finite collection $E_1, E_2, ..., E_n$ of pairwise disjoint sets in $\mathcal{F}$,

   $$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i).$$

   Proof. Using Axiom (ii) and considering the events $E_{n+1}, E_{n+2}, ...$ as empty sets, we have

   $$P\left(\bigcup_{i=1}^{\infty} E_i\right) = P\left(\bigcup_{i=1}^{n} E_i\right). \tag{1}$$

   Also, we have

   $$\begin{aligned} \sum_{i=1}^{\infty} P(E_i) &= \lim_{m\to\infty} \sum_{i=1}^{m} P(E_i) \\ &= \sum_{i=1}^{n} P(E_i) + \lim_{m\to\infty} \sum_{i=n+1}^{m} P(E_i) \\ &= \sum_{i=1}^{n} P(E_i) + \lim_{m\to\infty} 0 \\ &= \sum_{i=1}^{n} P(E_i). \end{aligned} \tag{2}$$

   The result can now be obtained using Axiom (ii) and (1)-(2).

3. P is a monotone function, i.e., if $E_1, E_2 \in \mathcal{F}$ and $E_1 \subset E_2$, then $P(E_1) \leq P(E_2)$.

   Proof. We can write E_2 as

$$E_2 = E_1 \cup (E_2 - E_1).$$

   This implies that

$$P(E_2) = P(E_1) + P(E_2 - E_1) \text{ [using Axiom (ii)]}. \tag{3}$$

   Now since $P(E_2 - E_1) \geq 0$ [using Axiom (i)], $P(E_2) \geq P(E_1)$.

4. For any $E \in \mathcal{F}$, $0 \leq P(E) \leq 1$.

   Proof. Since $E \subset \Omega$, $P(E) \leq P(\Omega) = 1$. Also $P(E) \geq 0$ from Axiom (1). This proves the result.

5. $P(E^C) = 1 - P(E)$

   Proof. Since $E \cup E^C = \Omega$, we have $P(E \cup E^C) = P(\Omega)$. This implies that $P(E) + P(E^C) = 1$ and the result follows.

6. **Law of Addition:** If $E_1, E_2 \in \mathcal{F}$, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2).$$

   Proof. We can write

$$E_1 \cup E_2 = E_1 \cup (E_2 - (E_1 \cap E_2)). \tag{4}$$

   This implies that

$$P(E_1 \cup E_2) = P(E_1) + P(E_2 - (E_1 \cap E_2)). \tag{5}$$

   Also from (3), we have

$$P(E_2 - E_1) = P(E_2) - P(E_1), \text{ if } E_1 \subset E_2.$$

   Here since $(E_1 \cap E_2) \subset E_2$, therefore

$$P(E_2 - (E_1 \cap E_2)) = P(E_2) - P(E_1 \cap E_2). \tag{6}$$

The result now follows by substituting (6) in (5).

7. If $E_1, E_2, E_3 \in \mathcal{F}$, then

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_2 \cap E_3) \\ - P(E_3 \cap E_1) + P(E_1 \cap E_2 \cap E_3).$$

8. **Principle of Inclusion-Exclusion:** If $E_1, E_2, ..., E_n \in \mathcal{F}$, then

$$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i) - \sum_{i<j}^{n} P(E_i \cap E_j) + \sum_{i<j<k}^{n} P(E_i \cap E_j \cap E_k) \\ - ... + (-1)^n P\left(\bigcap_{i=1}^{n} E_i\right).$$

MTL106: Probability and Stochastic Processes

(Lecture 2)

January 6, 2026

Axiomatic Approach of Probability: In the modern (axiomatic) approach of probability theory, we do not concerned about how probabilities are assigned to some events in the sample space Ω. We define the probability for certain types of subsets of the sample space Ω. For this, we use some axioms that are consistent with the three properties of the classical as well as relative frequency approaches. The axiomatic probability approach is based on some concepts of the set theory and measure theory. We first define these concepts and then move to the axiomatic definition of probability.

Class of Sets: A set whose elements are themselves set is known as a class of sets. Eg., $\mathcal{A} = \{\{1\}, \{1, 2\}, \{2, 3, 4\}\}$ is a class of sets.

Set Function: Let $\mathcal{C}$ be a class of sets. A function $\mu : \mathcal{C} \to \mathbb{R}$ is known as a *set function*. This means a set function is a real valued function whose domain is a class of sets.

Remark 1: Axiomatic approach deals with the probability assignments to the members of an appropriately chosen class $\mathcal{C}$ of subsets of the sample space Ω. Eg., (1) if $\Omega = \mathbb{R}$, then $\mathcal{C}$ may be a class of all open intervals in $\mathbb{R}$, (2) If Ω is a countable set, then $\mathcal{C}$ may be a class of all singletons $\{\omega\}$, $\omega \in \Omega$.

Remark 2: The members of the class of sets $\mathcal{C}$ are known as *basic sets*.

Event Space and Events: Suppose $\mathcal{F}$ is a class of sets for which the probability assignment to be done. Then the class $\mathcal{F}$ is known as the *event space* and elements of $\mathcal{F}$ are called *events*. Also $\mathcal{F}$ satisfies the following properties:

1. $\Omega \in \mathcal{F}$,
2. $E \in \mathcal{F} \Rightarrow E^C \in \mathcal{F}$,
3. $E_j \in \mathcal{F} \Rightarrow \bigcup_{j=1}^{\infty} E_j \in \mathcal{F}$, $j = 1, 2, \ldots$

σ-Field: A σ-field of subsets of sample space Ω is a class $\mathcal{F}$ of subsets of Ω which satisfies the following properties:

1. $\Omega \in \mathcal{F}$,
2. $E \in \mathcal{F} \Rightarrow E^C \in \mathcal{F}$,

3. $E_j \in \mathcal{F} \Rightarrow \bigcup_{j=1}^{\infty} E_j \in \mathcal{F},\ j = 1, 2, \ldots$

It is clear that a σ-field is a non-empty class of subsets of Ω that is closed under the formation of countable unions and complements and contains the null set ϕ.

Remark 3: The event space is basically a σ-field.

Remark 4: $\mathcal{F} = \{\phi, \Omega\}$ is known as a *trivial* σ-field.

Remark 5: $\mathcal{F} = \{E, E^C, \phi, \Omega\}$ is the *smallest* σ-field containing the set E.

Remark 6: The couplet $(\Omega, \mathcal{F})$ is known as a *measurable space.*

Example 1: Let $\Omega = \{A, B, C\}$. Then one possible σ-field is

$$\mathcal{F} = \{\phi, \{A\}, \{B, C\}, \{A, B, C\}\}.$$

Example 2: It might be possible that a sample space can be a set of real numbers in some situations where outcomes are continuous and can take any value within a certain range. Eg., measurements of temperature, height, weight, distance, blood pressure, sugar level, air quality index, etc. are continuous and take values in an interval. If $\Omega = \mathbb{R}$, then $\mathcal{F}$ might be ϕ, $\mathbb{R}$, any countable union of open intervals within $\mathbb{R}$, any countable intersection of open intervals within $\mathbb{R}$, complements of above sets within $\mathbb{R}$, etc.

Probability Function: Let $\mathcal{F}$ be a σ-field of subsets of Ω. A *probability function* (or a *probability measure*) is a set function $P : \mathcal{F} \to \mathbb{R}$ and satisfies the following three axioms:

1. $P(E) \geq 0 \quad \forall \quad E \in \mathcal{F}$,
2. $P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$, where $E_1, E_2, \ldots$ is a countably infinite collection of mutually exclusive events ($E_i \cap E_j = \phi,\ i \neq j,\ i, j = 1, 2, \ldots$),
3. $P(\Omega) = 1$.

Remark 5: The triplet $(\Omega, \mathcal{F}, P)$ is known as a probability space.

MTL106: Probability and Stochastic Processes

Lecture 1

January 2, 2026

The Need of Probability as a Mathematical Tool

In our day to day life, we come across a lot of uncertainties. Some examples are provided below:

- Predicting quarterly revenue of an organization.
- Number of customers likely to churn before next quarter.
- Demand for a product at a future time period.
- Number of views for an Instagram reel or a YouTube video.
- Outcome of the cricket matches between India and Australia.

The Need of Probability as a Mathematical Tool (Contd.)

- It is essential to develop a mathematical tool that can be used to predict and measure such uncertainties.
- Probability is a fundamental mathematical tool to understand and model the uncertainty associated with an event.

Random Experiment

A Random Experiment is an experiment in which:

- All outcomes of the experiment are known in advance.
- Any performance of the experiment results in an outcome that is not known in advance.
- The experiment can be repeated under identical conditions.

In probability theory we study this uncertainty of a random experiment.

Sample Space

The collection of all possible outcomes of a random experiment is called the Sample Space. Usually a sample space is denoted by Ω.

- Outcome of a cricket match: $\Omega = \{\text{Win, Lose, Draw}\}$.
- Customer Churn: $\Omega = \{\text{Churn, No Churn}\}$.
- Analyzing YouTube views: $\Omega = \{0, 1, 2, \ldots, 1000, \ldots, 10000, \ldots\}$.
- Lifetime (T) of a new product: $\Omega = \{T \mid 0 < T < \infty\}$.

Event

An Event 'E' is a subset of the sample space Ω ($E \subseteq \Omega$) and we calculate probability with respect to the event E. So if the outcome of a random experiment is a member of the set E, one can say that the event E has occurred.

- Customer churn (C) from a customer base of 100 customers:
  $\Omega = \{C \mid 0 \leq C \in \mathbb{Z} \leq 100\}$.
  $E_1 = \{C \mid C < 20\}$.
  $E_2 = \{C \mid C > 40\}$.
  $E_3 = \{C \mid 20 < C < 40\}$.
- Outcome of a cricket match:
  $\Omega = \{\text{Win, Lose, Draw}\}$.
  $E_1 = \{\text{Win}\}$.
  $E_2 = \{\text{Lose}\}$.
  $E_3 = \{\text{Draw}\}$.

Event (Contd.)

Remark 1. Two events E_1 and E_2 are said to be Mutually Exclusive, if they cannot occur simultaneously, i.e., $E_1 \cap E_2 = \varnothing$, the empty set.

- If two competing companies are bidding for the same contract, only one can win. If company A wins, Company B cannot, and vice versa.
- The outcome of a surgery could be either successful or unsuccessful. These outcomes are mutually exclusive.

Remark 2. Equally Likely events are those that have the same probability of occurring.

- Lottery for holiday leave.
- Random distribution of promotional coupons.
- Random customer survey selection.

Assignment of Probabilities

It might be possible that in a random experiment some events are more likely to occur than the others. See following examples:

- In a match between a top-ranked team and a lower-ranked team, the higher-ranked team is more likely to win.
- A person with a high credit score is more likely to get a loan approved than someone with a low credit score.

Assignment of Probabilities (Contd.)

- It is desirable to measure the likelihoods of the occurrences of various events.
- Probability of an event is a quantitative measure of uncertainty with which that event occurs.
- We assign a real number $P(E) \in [0, 1]$ to each event.
- We must have $P(\Omega) = 1$.

Classical Approach

- The Classical Approach of probability assignment is used when there are finite number of equally likely outcomes for some random experiment.
- Suppose $\Omega = \{\omega_1, \omega_2, \ldots, \omega_k\}$ represents a finite sample space with $k(\in \mathbb{N})$ possible outcomes.
- Let $E \subseteq \Omega$ is an event and $n(E)$ denotes the number of elements in E.
- Let $\omega \in \Omega$ is favorable to an event E, i.e., $\omega \in E$.
- According to the classical approach, the probability of the occurrence of an event E is the ratio of the *number of favorable outcomes* to E and *the total number of outcomes*, i.e.,

$$P(E) = \frac{n(E)}{n(\Omega)} = \frac{n(E)}{k}.$$

Classical Approach (Contd.)

For an event E, the probabilities assigned through the classical approach satisfy the following conditions:

1. $P(E) \geq 0$.
2. $P\left(\bigcup_{i=1}^{k} E_i\right) = \sum_{i=1}^{k} P(E_i)$, where $E_1, E_2, \ldots, E_k$ are mutually exclusive events ($E_i \cap E_j = \varnothing,\ i, j \in \{1, 2, \ldots, k\},\ i \neq j$).
3. $P(\Omega) = 1$.

Remark 3. The classical approach of probability assignment is confined to those random experiments only which have a finite number of equally likely outcomes. This makes the use of the classical approach very restricted.

Relative Frequency Approach

- Suppose we repeat a random experiment under identical conditions such that the trials are independent.
- Let $f_N(E)$ denotes the frequency of an event E in N independent trials.
- Let $r_N(E) = \frac{f_N(E)}{N}$ represents the corresponding relative frequency.
- Then according to the Relative Frequency Approach, probability of the event E is given by

$$P(E) = \lim_{N \to \infty} r_N(E) = \frac{f_N(E)}{N}.$$

- Note that the experiment is repeated a large number (N) of times and the approximation $P(E) \approx r_N(E)$ is used for assigning probability to the event E.

Relative Frequency Approach (Contd.)

For an event E, the probabilities assigned through the relative frequency approach also satisfy the following conditions:

1. $P(E) \geq 0$.
2. $P\left(\bigcup_{i=1}^{k} E_i\right) = \sum_{i=1}^{k} P(E_i)$, where $E_1, E_2, \ldots, E_k$ are mutually exclusive events ($E_i \cap E_j = \varnothing,\ i, j \in \{1, 2, \ldots, k\},\ i \neq j$).
3. $P(\Omega) = 1$.

Remark 4. Since the relative frequency approach is based on an approximation, it might not be precise enough. Also due to various logistical constraints, it is not always feasible to repeat the random experiment a large number of times. These are the limitations of the relative frequency approach.